

AI, Edge, and Cloud Architecture

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Product: mooreVIEW
Audience: Integrators, cloud operators, IoT-Link field deployment
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One-line summary

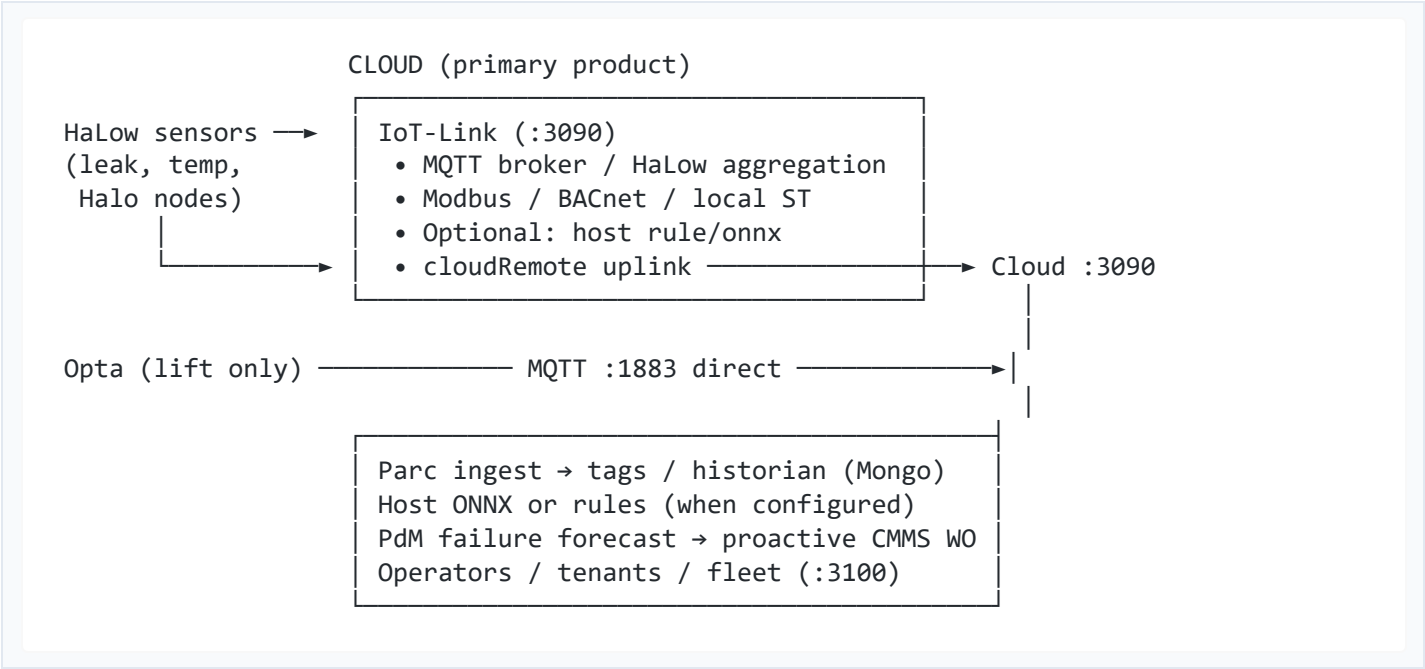
Cloud is the product (fleet, PdM, CMMS, historians, host ML). **IoT-Link is the production edge** (HaLow hub, concentrated I/O, optional edge AI). **est-pc / MVP Suite PC is dev/eval only**. Remote Opta lift panels are cloud-connected leaves.

1. Layer roles — do not conflate

Layer	Deploy target	Role
Cloud SaaS	Droplet :3100 + runtime :3090 + Mosquitto	Multi-tenant ops, Parc ingest, host ONNX/rules, PdM forecast, CMMS, fleet
IoT-Link	Compulab Linux gateway (deploy/iot-link/)	Production appliance: Modbus/BACnet, local MQTT hub, HaLow aggregation, cloudRemote uplink
est-pc / MVP Suite	Windows/Linux dev PC	Full-stack evaluation, sim studies, firmware parity — not the field appliance story
Opta (remote lift)	Arduino panel at store	Grease/lift I/O, lightweight pseudo-AI, MQTT Parc direct to cloud

Appliances **link to cloud**; they do not replace it. Field buses (Modbus RTU, BACnet/IP, ONVIF discover) stay on the edge per docs/EST_PC_PARITY.md .

2. Data flow (reference architecture)



Circle K phase 1 (lift only): Opta → cloud Mosquitto — no IoT-Link required.
Circle K phase 2 (HVAC-R, leaks, coolers): IoT-Link at store → **cloudRemote** → same cloud tenant.

3. AI vs pseudo-AI — where each resides

MooreVIEW uses **four processing layers**. Only **host ONNX** is real neural inference today.

3.1 Device pseudo-AI (Arduino Opta firmware)

Item	Detail
Code	firmware/arduino-opta-mqtt-st/MooreviewOptaMqttSt/mv_edge_ai.cpp
Pseudo MCSA	mcsa[] spectra synthesized from CT amps — not a real FFT
edgeAi[]	Rule-based labels on pump motor start (startMs , amp ratio)
Labels	healthy, seal_leak, clog_ragging, impeller_worn
Uplink	Parc telemetry → cloud or via IoT-Link LAN broker

This is **heuristic scoring**, not neural inference. Suitable for start detection and coarse health at the panel.

3.2 Device true MCSA (Nexcomm MCXN947 class)

Item	Detail
Template	mcxn947_hvac_mcsa, lift MCSA platforms

Signal	Real FFT cooked spectra in <code>mcsa[]</code> (fund, rotor, bearing bins)
Classification	Typically on MooreVIEW host after uplink
Typical site	C-store RTU, IoT-Link + HaLow/MCSA node

Signal processing on device; **decision layer** on cloud or IoT-Link runtime.

3.3 Host inference (rules + ONNX)

Item	Detail
Code	<code>src/inference/hostInference.js</code> , triggered from <code>src/parc/mqttCentralHub.js</code>
Runs where	Any process hosting the MQTT Parc hub — cloud :3090 or IoT-Link :3090
Storage	MongoDB <code>edge_inference_ts</code> (when logger configured)

Backends (`settings.inference.backend`):

Backend	Type	Notes
<code>rule</code>	Pseudo-AI	Thresholds on MCSA metrics + start context (<code>ruleClassifier.js</code>)
<code>onnx</code>	Real ML	Requires model in <code>data/models/</code> ; uses <code>onnxBackend.js</code>
<code>auto</code>	Prefer ONNX, fall back to rules	Default path

Modes (`settings.inference.mode`):

Mode	Behavior
<code>host-supplement</code>	Run host only if device did not send <code>edgeAi</code> — production default
<code>host-override</code>	Always run host (ignore device <code>edgeAi</code>)
<code>host-on-start</code>	Host on motor starts when no device <code>edgeAi</code>
<code>off</code>	No host inference

3.4 PdM failure forecast (always host / cloud)

Item	Detail
Code	<code>src/pdm/failureForecast.js</code> , <code>src/pdm/featureAlign.js</code>
Type	Statistical regression on historian — not a neural net
Inputs	SCADA pens (<code>MOTOR_START_MS</code> , run amps) + aligned <code>edge_inference_ts</code> windows
Methods	<code>health_index</code> , <code>run_amps_creep</code> , <code>start_time_ms</code> , <code>starts_analytics</code>
Output	RUL estimate, proactive CMMS work orders

Belongs at **cloud scale** (or IoT-Link with Mongo for isolated sites).

3.5 Camera inference (separate path)

Item	Detail
Code	src/cameras/cameraInference.js
Runs where	Edge with camera access (IoT-Link); results uplinked or stored locally
Backends	HTTP to external service, or stub (dev only)
Storage	MongoDB edge_inference_ts, camera_events

Unrelated to lift MCSA / Parc pump paths.

4. Recommended placement by scenario

Scenario	Edge (IoT-Link / Opta)	Cloud
Circle K lift only (Opta)	Opta: pseudo edgeAi + mcsa-lite	Host ONNX + PdM + CMMS
C-store full (HaLow + RTU + lift)	IoT-Link: buses, HaLow hub; optional local host	Fleet PdM, historian, operators
Putnam-style Nexcomm lift	None (cellular MQTT direct)	Cloud ingest + lift_station_epi
Camera + leak detection	IoT-Link: ONVIF + HaLow peers	Site agent + cloud live view

HaLow sensor nodes: telemetry only → IoT-Link MQTT broker → cloud. No heavy AI on the HaLow MCU.

Edge AI when appropriate: motor starts / MCSA at IoT-Link or Opta; **fleet learning, WO policy, and county-scale ops** stay in cloud.

5. PdM study matrix (6-month lift station eval)

From docs/testing/LIFT-STATION-6MO-PDM-STUDY-TEST.md — validates paths in **est-pc** sim:

Scenario ID	MCSA source	Host path	Early warning
opta-mcsa-regression	Opta pseudo MCSA	None (PdM regression only)	Late (month 6+)
opta-mcsa-onnx	Opta pseudo MCSA	Host ONNX	Early (month 1)
true-mcsa-regression	True FFT MCSA (sim)	Host rules	Early
true-mcsa-onnx	True FFT MCSA (sim)	Host ONNX	Early

Production recommendation for cloud lift fleets: uplink pseudo or true MCSA → **enable host ONNX on cloud :3090** → PdM → proactive CMMS WO.

Default production profile: inference.hostEnabled: true , mode: host-supplement , backend: auto or onnx .

6. What belongs where — quick reference

Capability	Cloud	IoT-Link	Opta	est-pc PC (dev)
Multi-tenant orgs / users	✓	—	—	eval
MQTT Parc fleet ingest	✓	uplink	direct	✓
Modbus / BACnet LAN	—	✓	—	✓
HaLow sensor aggregation	—	✓	—	sim
Pseudo MCSA + edgeAi	—	—	✓	✓
True FFT MCSA	via device	via device	—	sim
Host rule / ONNX inference	✓	optional	—	✓
PdM failure forecast	✓	w/ Mongo	—	✓
Proactive CMMS WO	✓	w/ CMMS	—	✓
ONVIF cameras	via agent	✓	—	✓
4-way PdM sim studies	—	—	—	✓

7. est-pc vs production deploy artifacts

Artifact	Purpose
est-pc repo / MVP Suite PC	Development, training, automated PdM studies, project generation
deploy/cloud/debian/install-saas.sh	Production cloud droplet
deploy/iot-link/install-generic.sh	Production IoT-Link edge image
deploy/cloud/.env.cstore-opta-parc.example	Cloud runtime env for Opta Parc fleets
deploy/iot-link/.env.generic.example	Blank IoT-Link appliance

Do **not** position the Windows MVP Suite PC as the primary field appliance. **IoT-Link** is the concentrated I/O product; **cloud** is the operator and analytics plane.

8. Related documentation

Topic	Path
Edge vs cloud parity	docs/EST_PC_PARITY.md
Cloud operator guide	docs/CLOUD_USER_GUIDE.md
C-store / Circle K cloud onboarding	docs/CSTORE_OPTA_PARC_CLOUD.md

Opta direct to cloud	docs/OPTA_PARC_CLOUD.md
IoT-Link install	deploy/iot-link/README-generic.md
C-store HaLow + IoT-Link story	docs/nexcomm/cstore-halow-story.md
Lift PdM study acceptance	docs/testing/LIFT-STATION-6M0-PDM-STUDY-TEST.md
Host inference settings	src/settings/inferenceSettings.js

9. Key source files (implementers)

Concern	Path
Opta pseudo-AI / MCSA-lite	firmware/.../mv_edge_ai.cpp
Host inference orchestration	src/inference/hostInference.js
Rule classifier (pseudo)	src/inference/ruleClassifier.js
ONNX backend	src/inference/onnxBackend.js
Parc hub → host trigger	src/parc/mqttCentralHub.js
PdM forecast	src/pdm/failureForecast.js
Feature alignment	src/pdm/featureAlign.js
Cloud remote uplink	src/integrations/applianceCloudRelay.js
IoT-Link service	deploy/iot-link/mooreview-iot-link-generic.service